Definition:

Cyclic group:

A group $(G, *)$ is said to be cyclic, if there exists an element $a \in G$ such that every element x of G can be expressed as $x = a^n$ for some integer n.

In such a case, the cyclic group is said to be generated by a or a is a generator of G. G is also denoted by $\{a\}$.

For example,

① if $G = \{1, -1, i, -i\}$ (finite group)

then $(G, \times)$ is a cyclic group with the generator i, for

$1 = i^4$, $-1 = i^2$, $i = i^1$ and $-i = i^3$.

For this cyclic group, $-i$ is also a generator.

$\therefore$ For any generator the inverse also a generator.

$\therefore G = \{i\}$ or $\{-i\}$

$\therefore G = \{i\}$ is a finite cyclic group.

(2) Let $Z = \{0, \pm 1, \pm 2 \ldots\}$ be the infinite group w.r.t. '+' for addition $x = na$

Here -1 & $+1$ are Generators of Z.

$\therefore (Z, +)$ is an infinite cyclic group.

Note:

Every Subgroup of a cyclic group is cyclic.

Ex:

Let $G = \{1, -1, i, -i\}$ be the cyclic group.

& let $H = \{1, -1\}$ be the subgroup of G.

Since -1 is a generator of H,

$\therefore$ H is cyclic.

① P.T every cyclic group is abelian but the Converse is not true.

proof:

Let G be a cyclic group.

$\therefore$ for $p, q \in G$ such that $p = a^r$ & $q = a^s$ for some $r, s \in \mathbb{Z}$ (integers)

$$\therefore a^r * a^s = a^{r+s} = a^s * a^r$$

$$\therefore p * q = q * p$$

$\therefore$ G is abelian.

Conversely,

Let $(R, +)$ be the abelian group.

Since it has no generators,

$\therefore$ $\mathbb{R}$ is not cyclic.

$\therefore$ Converse is not true.

② P.T. every Subgroup of a cyclic group is abelian.

proof:

Let G be a cyclic group.

$\therefore$ $G = \{a\}$ for $a \in G$.

Let $H \subseteq G$ be the Subgroup,

To prove: H is cyclic

Let m be the least +ve integer such that $a^m \in G$

To prove : $H = \{a^m\}$

Let $x \in H \Rightarrow x \in G$

$\Rightarrow x = a^n$ but $n > m$

$\therefore n = mq + r, \quad 0 \leq r < m$

Since m is the least +ve integer

$\therefore r = 0$

$n = mq$

$x = a^{mq} = (a^m)^q$

$\therefore H = \{a^m\}$

$\therefore H$ is cyclic

$\therefore H$ is abelian.

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③ P.T. every group of order prime is cyclic.

proof: Let G be a group.

s.t. $O(G) = p$ (prime)

Let $a \in G$ s.t. $a \neq e$

& let $H = \{a\} \subseteq G$ be the subgroup

s.t $O(H) = m < p$

$\therefore O(H) / O(G)$ (by lagrange's theorem)

$O(H) = 1$ (or) p

Since $a \neq e$

$\therefore O(H) \neq 1$.

$\therefore O(H) = p = O(G)$

$\therefore H$ is cyclic & It is abelian

$\therefore G$ is cyclic.

(4) P.T. center of a group is a subgroup of G.

proof:

Let $Z = \{ a \in G \mid a * x = x * a$ for all $x \in G \}$

be the center of the group.

To prove: Z be the subgroup of G.

ie for $a, b \in Z$, $a * b^{-1} \in Z$

Since $a, b \in Z$,

$\therefore a * x = x * a$

& $b * x = x * b$ for all $x \in G$.

TO prove: $a * b^{-1} \in Z$ for $a, b \in Z$

ie $(a * b^{-1}) * x = x * (a * b^{-1})$

for $a, b \in Z$

LHS:

$$(a * b^{-1}) * x = a * (b^{-1} * x)$$

$$= a * ((x^{-1} * b)^{-1})$$

$$= a * ((b * x^{-1})^{-1})$$

$$= a * (x * b^{-1})$$

$$= (a * x) * b^{-1}$$

$$= (x * a) * b^{-1}$$

$$= x * (a * b^{-1}) = \text{RHS}.$$

$\therefore a * b^{-1} \in Z$ for $a, b \in Z$

$\therefore Z$ is a subgroup.

Note:

Every group of prime order is cyclic & every cyclic is abelian.

P.T. $(Z_4, +_4)$ is an abelian group.

* last page

[0] = {.... −8 −4, 0, 4, 8 ....}
[1] = {.... −7 −3, 1, 5, 9, ...}
[2] = {.... −6, −2, 2, 6, 10, ...}
[3] = {...., −5, −1, 3, 7, 11, ...}

Sol:

Let $Z_4 = \{[0], [1], [2], [3]\}$

Cayley's table

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Since none of the element in any rows & any columns are repeated

∴ each & every element posses unique inverse in Z_4.

∴ $(Z_4, +_4)$ is an abelian group.

Note:

$(\mathbb{Z}_n, +_n)$ is an abelian group.

proof. refer the text book.

(2) Find the order of each element of the groups.

i) $G = \{1, -1, i, -i\}$

ii) $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$

i) $o(-1) = 2$ since $(-1)^2 = 1$

$o(1) = 1$ $(1)^1 = 1$

$o(i) = 4$ $(i)^4 = 1$

$o(-i) = 4$ $(-i)^4 = 1$.

(ii) $o([0]) = 0$

$o([1]) = 4 \Rightarrow [1] +_4 [1] +_4 [1] +_4 [1]$

$= 0$

$o([2]) = 2$ since $[2] +_4 [2] = [0]$

$o([3]) = 4$ since $[3] +_4 [3] +_4 [3] +_4 [3]$

$= [0]$

Note:

$(\mathbb{Z}_p - \{[0]\}, \cdot_p)$ is a group whenever p is a prime.